

MINDPILOT BUSINESS PLAN

OmniPilot Hardware

From Voice Assistant to Assistive AI Platform

Prepared for: Soumya Swain | Sidewall

Date: April 17, 2026

Version: v1.0.0 | Confidential

Table of Contents

Phase 1: Strategy

1.1 The Hardware Vision	4
1.2 Three Hardware Verticals	4
1.3 Why Blind Assistive is #1 Priority	5
1.4 Market Sizing	6
1.5 Risks & Mitigation	7

Phase 2: Business Design

2.1 Revenue Model	8
2.2 Unit Economics	9
2.3 Manufacturing: India vs Shenzhen	9
2.4 Certification Requirements	10
2.5 Distribution Channels	10
2.6 Funding & Grants	11

Phase 3: Product Design

3.1 Blind Assistive Device Architecture	12
3.2 BLE Pendant (Omi Competitor)	13
3.3 Drone Voice Integration	14
3.4 Smart Home (Software-Only)	15
3.5 Build Path & Roadmap	15

Phase 4: Execution

4.1 90-Day Action Plan	16
4.2 Budget Allocation	17
4.3 Key Partnerships	17
4.4 Success Metrics	18

About This Document

This business plan follows the **MindPilot 4-Phase Framework**: Strategy (Why & Where) → Business Design (How It Works) → Product Design (What We Build) → Execution (Getting It Done). Each phase builds on the previous, ensuring no critical dimension is skipped.

PHASE 1

Strategy – The "Why & Where"

1.1 The Hardware Vision

OmniPilot started as an AI voice assistant for productivity — a software companion that helps you think, remember, and act. But voice AI becomes transformative when paired with purpose-built hardware. A \$10 BLE pendant streaming audio to your phone. A camera-equipped device guiding a blind person through a busy Indian street. A voice module letting a farmer command a crop-dusting drone in Hindi.

Core Insight

Software AI is a crowded race to the bottom. Hardware AI creates **defensible moats** — certification barriers, supply chain relationships, form factor IP, and distribution partnerships that software-only players cannot replicate overnight.

The hardware extension turns OmniPilot from a voice app into an **assistive AI platform** — one that can serve the visually impaired, agriculture, wearable computing, and smart home markets through a unified AI brain with specialized peripherals.

1.2 Three Hardware Verticals

Vertical	Product	Price Range	Priority
Blind Assistive AI	Camera + voice pendant for visually impaired	Rs 5,000 – 13,000	#1 — Highest
Drone Voice Control	Voice command module for agriculture drones	B2B licensing	#2 — Medium
BLE Pendant	Always-on audio companion (Omi competitor)	Rs 5,000 – 7,000	#3 — Steady

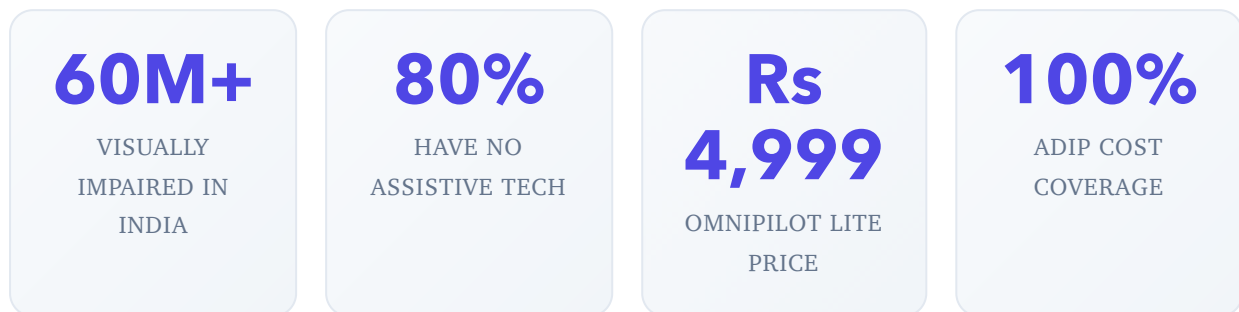
Analogy

Think of OmniPilot as the brain (AI engine) and each hardware vertical as a body. The blind assistive device gives it eyes. The drone module gives it wings. The pendant gives it ears. Same brain, different superpowers.

[Chart missing: system_architecture]

Fig 1.1 — OmniPilot unified AI architecture across hardware verticals

1.3 Why Blind Assistive is #1 Priority



India has the world's largest visually impaired population. The existing assistive devices — OrCam MyEye (Rs 2.5L), Envision Glasses (Rs 3.5L) — are priced for Western markets. OmniPilot Lite at Rs 4,999 is **50x cheaper**.

ADIP Scheme – The Game Changer

The Government of India's **ADIP (Assistance to Disabled Persons) scheme** covers 100% of assistive device costs under Rs 15,000. OmniPilot Lite at Rs 4,999 qualifies automatically. This means **zero cost to the end user** — the government pays. Tenders are placed through NGOs for 500–5,000 units at a time.

Competitor Comparison

Feature	OrCam MyEye 3	Envision Glasses	OmniPilot Lite
Price	Rs 2,50,000	Rs 3,50,000	Rs 4,999
Hindi Support	Limited	None	Native Hindi + Regional
ADIP Eligible	No (too expensive)	No (too expensive)	Yes (100% covered)
Obstacle Detection	No	Basic	MiDaS depth + alerts
Text Reading	OCR (English)	OCR (English)	PaddleOCR (Hindi/English)
Scene Description	Basic labels	LLM-powered	LLM + context memory
Form Factor	Glasses clip	Smart glasses	Clip-on pendant

[Chart missing: hw_blind_market]

Fig 1.2 — Visually impaired population and assistive tech penetration in India

CONFIDENTIAL

OmniPilot Hardware | Phase 1: Strategy

1.4 Market Sizing



[Chart missing: hw_opportunity_map]

Fig 1.3 — OmniPilot hardware opportunity across verticals and price points

Market Breakdown by Vertical

Vertical	India Market (2026)	Growth Rate	OmniPilot Addressable
Blind Assistive Tech	\$890M	18% CAGR	\$120M (ADIP + D2C)
Agriculture Drones	\$1.2B	25% CAGR	\$45M (voice add-on)
AI Wearables	\$680M	32% CAGR	\$175M (consumer pendant)
Smart Home (SW)	\$3.5B	22% CAGR	\$50M (voice control)

[Chart missing: portfolio_fit]

Fig 1.4 — OmniPilot portfolio fit across software and hardware products

1.5 Risks & Mitigation

Risk	Severity	Mitigation
ADIP scheme changes or budget cuts	High	Diversify into D2C + Amazon; keep price under Rs 15K for any future scheme threshold
Hardware quality at low BOM	Medium	Use proven SoCs (nRF52840, ESP32-S3); partner with VVDN for manufacturing QA
LLM API cost per user per month	Medium	On-device inference for 80% of tasks; cloud LLM only for complex scene description
Competitors enter India market	Low-Med	First-mover on Hindi/regional; ADIP certification moat; NGO distribution lock-in

BIS/WPC certification
delays

Medium

Start applications in Month 1; use pre-certified BLE
modules (nRF52840 DK)

PHASE 2

Business Design – The "How It Works"

2.1 Revenue Model

Channel	Vertical	Pricing	Volume (Year 1)
ADIP Government Tenders	Blind Assistive	Rs 4,999–12,999/unit	2,000–5,000 units
Amazon / Flipkart D2C	Blind Assistive + Pendant	Rs 5,999–7,499	500–1,000 units
NGO Partnerships	Blind Assistive	Bulk at Rs 4,499	1,000–3,000 units
B2B Licensing	Drone Voice	Rs 15,000/license + royalty	50–200 licenses
GeM Portal	Blind Assistive	Rs 4,999–12,999	500–2,000 units

Revenue Projection (Year 1)

Conservative: Rs 1.5 Cr (3,000 blind assistive units via ADIP + 500 pendants)

Optimistic: Rs 4.2 Cr (8,000 units across all channels + drone licensing)

Revenue by Quarter (Year 1)

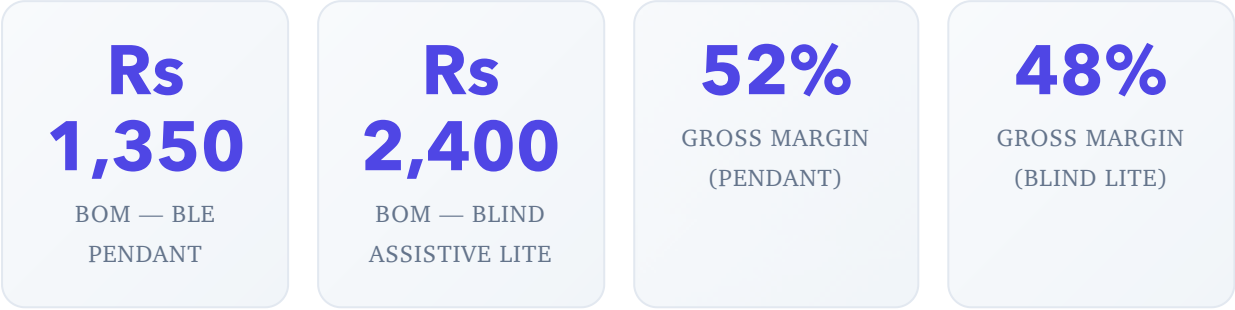
Quarter	Blind Assistive	Pendant	Drone	Total
Q1	Rs 0	Rs 0	Rs 0	Rs 0 (build phase)
Q2	Rs 25L (500 units)	Rs 3.5L (50 units)	Rs 0	Rs 28.5L
Q3	Rs 50L (1,000 units)	Rs 14L (200 units)	Rs 7.5L (50 lic)	Rs 71.5L

Q4	Rs 75L (1,500 units)	Rs 17.5L (250 units)	Rs 15L (100 lic)	Rs 107.5L
----	----------------------	----------------------	------------------	-----------

[Chart missing: data_flow]

Fig 2.1 — Data flow from device through AI pipeline to user interaction

2.2 Unit Economics



[Chart missing: hw_bom_comparison]

Fig 2.2 — Bill of Materials comparison across product tiers

BOM Breakdown – BLE Pendant

Component	Spec	Unit Cost
MCU	nRF52840 (BLE 5.0, ARM Cortex-M4)	Rs 250
Microphone	MEMS (ICS-43434 or equiv.)	Rs 60
Speaker	13mm driver + amplifier	Rs 80
Battery	200mAh LiPo + charge IC	Rs 120
PCB + Assembly	4-layer, SMT assembly	Rs 180
Enclosure	Injection-molded ABS, clip mount	Rs 150
Misc (antenna, USB-C, packaging)	Connectors, box, manual	Rs 160
Total COGS		Rs 1,000–1,350

2.3 Manufacturing: India vs Shenzhen

[Chart missing: hw_mfg_cost]

Fig 2.3 — Manufacturing cost comparison: India vs Shenzhen (with 22% import duty)

India Wins for Sub-5K Units

For volumes under 5,000 units, Indian manufacturing is **cheaper after 22% import duty** on China-imported electronics. Partners like VVDN Technologies (Gurugram), SFO Technologies (Kochi), and Optiemus Electronics (Noida) can handle full PCBA + assembly + testing.

Factor	India (VVDN/SFO)	Shenzhen
MOQ	500–1,000 units	3,000–5,000 units
Lead Time	6–8 weeks	4–6 weeks
Import Duty	0%	22% on electronics
BIS/WPC Ease	Easier (domestic)	Requires import license
Communication	Same timezone, English	WeChat, timezone gap
Quality at low vol	Good (ISO-certified)	Variable at low MOQ

2.4 Certification Requirements

Certification	Authority	Cost	Timeline	Required For
BIS (CRS)	Bureau of Indian Standards	Rs 2–3L	8–12 weeks	All electronic products sold in India
WPC (ETA)	Wireless Planning Commission	Rs 1–2L	4–8 weeks	BLE/WiFi radio devices
EMC/EMI	NABL-accredited lab	Rs 1–2L	2–4 weeks	All electronic devices
ADIP Listing	Ministry of Social Justice	Nominal	6–12 weeks	Government procurement eligibility
Total Budget		Rs 4–8L	3–4 months	

Certification Tip

Start BIS + WPC applications in **Month 1** while prototyping. Use pre-certified BLE modules (like nRF52840 DK with existing WPC approval) to reduce certification scope and timeline.

2.5 Distribution Channels

Channel	Target Segment	Commission / Cost	Priority
ADIP / GeM Portal	Government institutions, NGOs	Tender-based, 0% commission	Highest
NGO Partnerships (NAB, Saksham)	Visually impaired communities	Bulk discount (10–15%)	High
Amazon.in / Flipkart	D2C consumers, families	15–20% commission	Medium
Optical shops / rehab centers	Walk-in visually impaired	20–25% margin	Low-Med
B2B direct (drone companies)	Drone manufacturers	Direct sales team	Medium

2.6 Funding & Grants

Source	Amount	Type	Fit
BIRAC BIG Grant	Rs 50 Lakh	Non-dilutive grant	Excellent — assistive tech
DST PRAYAS	Rs 10 Lakh	Prototype grant	Excellent — hardware MVP
Startup India Seed Fund	Rs 20 Lakh	Equity/loan	Good
Villgro / Social Alpha	Rs 25–50 Lakh	Impact investment	Excellent — social impact
NASSCOM Foundation	Rs 10–20 Lakh	Tech for Good grant	Excellent
Atal Innovation Mission	Rs 10 Lakh	Grant	Good

Funding Strategy

Apply for **BIRAC BIG + DST PRAYAS simultaneously** (non-competing). Combined Rs 60L covers prototyping, certification, and first 500-unit manufacturing run without equity dilution.

PHASE 3

Product Design – The "What We Build"

3.1 Blind Assistive Device Architecture

Hardware Stack

Component	Spec	Purpose
Camera	OV5640 (5MP, autofocus)	Scene capture, text reading, object detection
Microphone	MEMS array (ICS-43434)	Voice input, ambient sound awareness
Speaker	Bone conduction / earpiece	Audio output without blocking ambient sound
Processor	ESP32-S3 (on-device) + phone (heavy AI)	Edge inference + cloud AI offload
Connectivity	BLE 5.0 + optional WiFi	Phone tethering, cloud API access
Battery	500mAh LiPo	6–8 hours mixed use
Form Factor	Chest clip pendant (45g)	Hands-free, camera at chest level

AI Pipeline

Layer	Model	Runs On	Latency
Object Detection	YOLOv8n (quantized INT8)	Phone (on-device)	<50ms
Depth / Obstacles	MiDaS v2.1 Small	Phone (on-device)	<80ms
Text Reading (OCR)	PaddleOCR (Hindi + English)	Phone (on-device)	<200ms
Scene Description	LLM (GPT-4o mini / Gemini)	Cloud API	500ms–1.5s
Navigation	Google Maps + depth fusion	Phone + cloud	<300ms

Voice I/O	Whisper (STT) + TTS	Phone / cloud	<500ms
-----------	---------------------	---------------	--------

80/20 On-Device Rule

80% of daily tasks (obstacle alerts, text reading, object identification) run **entirely on-device** with zero latency and zero API cost. Only complex scene description and navigation use cloud LLMs. This keeps the monthly API cost under **Rs 50/user**.

Product Tiers

Tier	Features	Hardware	Price
Lite	Voice assistant + obstacle alerts + OCR	BLE pendant + phone mic/speaker	Rs 4,999
Standard	Lite + camera + scene description	Camera pendant + phone	Rs 12,999
Pro	Standard + navigation + depth	Stereo camera + haptic	Rs 24,999

3.2 BLE Pendant (Omi Competitor)

The BLE pendant is OmniPilot's consumer wearable — an always-on AI companion that clips to your collar and streams audio to your phone. Think of it as AirPods for your brain: it listens, remembers, summarizes, and acts on your behalf.

Specifications

Spec	Value
MCU	nRF52840 (64MHz ARM Cortex-M4, 256KB RAM)
Audio	MEMS mic + Opus codec @ 16kHz
Connectivity	BLE 5.0 (audio stream to phone app)
Battery	200mAh LiPo — 8–12 hours continuous
Form Factor	Clip / pendant, 12g, waterproof IP54
Charging	USB-C, 0–100% in 45 minutes
BOM Cost	Rs 500–800 (\$6–10)
Retail Price	Rs 5,000–7,000

Market Positioning

Omi (US) sells at \$89 (Rs 7,500) with English-only support. OmniPilot pendant at Rs 5,000 with **Hindi + 10 Indian languages**, plus integration with the OmniPilot AI brain (memory, tasks, calendar, smart home). Same hardware, better localization, lower price.

3.3 Drone Voice Integration

Indian agriculture is rapidly adopting drones for crop spraying, but pilots need extensive training to operate them. Voice control removes the training barrier: a farmer speaks in Hindi, and the drone follows.

Voice Command Pipeline

Stage	Technology	Example
Voice Input	Whisper / IndicWhisper STT	"10 meter upar jao" (go up 10m)
Intent Parsing	NLU classifier + slot filling	Intent: ASCEND, altitude: 10m
Command Translation	MAVLink protocol generator	MAV_CMD_NAV_TAKEOFF alt=10
Safety Validation	Geofence + whitelist + dead man's	Check: within boundary? Command allowed?
Execution	Drone autopilot (PX4/ArduPilot)	Execute validated MAVLink command

[Chart missing: hw_drone_market]

Fig 3.1 — India agriculture drone market and voice control opportunity

Safety-First Design

Voice-controlled drones require **triple safety layers**: (1) Geofence boundaries that cannot be overridden by voice, (2) Command whitelist — only pre-approved commands accepted, (3) Dead man's switch — drone hovers/lands if voice link drops for >5 seconds.

3.4 Smart Home (Software-Only, Phase 5)

Smart home integration is a **software-only play** — no custom hardware needed. OmniPilot's voice AI connects to existing smart home ecosystems through established protocols.

Integration	Coverage	Effort
Tuya API	80% of Indian smart home devices (WiFi bulbs, plugs, cameras)	2–3 weeks SDK integration
HomeKit / Matter	Apple ecosystem (Thread, WiFi)	4–6 weeks certification
Google Home	Google Nest ecosystem	2–3 weeks API integration
Alexa Skills Kit	Amazon Echo ecosystem	2–3 weeks skill development

Why Phase 5?

Smart home integration is **low effort, high retention** — once a user controls their home through OmniPilot, switching costs are high. But it is not a differentiator for initial adoption. Ship it after the core hardware products establish the user base.

3.5 Build Path & Roadmap

[Chart missing: hw_roadmap]

Fig 3.2 — OmniPilot hardware development roadmap across 5 phases

Phase	Timeline	Deliverable	Revenue Start
Phase 1: Android App	Month 1–2	Blind assistive app (software MVP)	Free beta
Phase 2: BLE Pendant	Month 2–4	Pendant prototype + beta testing	Month 5
Phase 3: Camera Pendant	Month 4–6	Blind assistive hardware device	Month 7

Phase 4: Drone Module	Month 6–9	Voice control integration kit	Month 10
Phase 5: Smart Home	Month 9–12	Tuya + HomeKit integrations	Month 10

PHASE 4

Execution – The "Getting It Done"

4.1 90-Day Action Plan

Month 1: Software MVP + Certifications

Week	Action	Owner	Output
W1	Android app scaffold + camera integration	Dev team	Basic camera-to-AI pipeline
W2	YOLOv8n + PaddleOCR on-device integration	ML engineer	Object detection + Hindi OCR working
W3	Voice I/O (Whisper STT + TTS)	Dev team	Voice-driven assistive features
W4	BIS/WPC applications filed + beta app release	Compliance + Dev	Apps submitted, 10 internal testers

Month 2: Hardware Prototype

Week	Action	Owner	Output
W5	nRF52840 DK + MEMS mic BLE audio streaming	HW engineer	Audio stream to phone app working
W6	PCB design (4-layer, pendant form factor)	HW engineer	Gerber files ready for PCBA
W7	VVDN engagement + NRE quote	Ops / Founder	Manufacturing partner confirmed
W8	3D-printed enclosure + first 10 prototypes	HW + Industrial design	Working prototypes for beta

Month 3: Beta + Validation

Week	Action	Owner	Output
W9	Beta deployment to 10 blind users (via NAB)	Product + NAB	Real-world usage data collection
W10	Iterate on AI accuracy + UX feedback	ML + Product	AI accuracy >90% for common objects
W11	ADIP listing application + NGO outreach	Ops / Founder	Application submitted + 3 NGO LOIs
W12	Production PCB order (500 units) + packaging	Ops / HW	First production batch in manufacturing

The "App First" Strategy

Month 1 builds a **software-only blind assistive app** using the phone's existing camera and mic. This validates AI accuracy and user experience **before spending a rupee on hardware**. If the app doesn't work well, we iterate on software. If it works, we know the hardware will amplify it.

4.2 Budget Allocation

Rs 25-40L

TOTAL BUDGET (FIRST 1000 UNITS)

Rs 60L

AVAILABLE VIA GRANTS

Category	Amount (Rs)	% of Total	Details
Hardware prototyping	5-8L	20%	Dev kits, PCB runs, 3D printing, 10 prototypes
Manufacturing (500 units)	8-12L	30%	PCBA, enclosure, assembly, testing (VVDN)
Certification (BIS/WPC/EMC)	4-8L	15%	Lab testing + filing fees
Software development	3-5L	12%	Android app, AI pipeline, cloud infrastructure
Marketing + distribution	3-4L	10%	Amazon listing, NGO events, demo units
Operations + legal	2-3L	8%	Company registration, IP filing, compliance
Contingency	2-3L	5%	Unforeseen costs, iteration buffer
Total	25-40L	100%	

4.3 Key Partnerships

Partner	Type	Value to OmniPilot	Status
National Association for the Blind (NAB)	NGO / Distribution	Beta users, ADIP channel, credibility	To approach
Saksham Trust	NGO / Distribution	Pan-India visually impaired network	To approach

VVDN Technologies	Manufacturing	Full PCBA + assembly + testing	To approach
Garuda Aerospace	Drone B2B	Voice module integration partner	To approach
IoTechWorld Avigation	Drone B2B	Agriculture drone voice control	To approach
IIT Delhi / IISc	Research	IndicWhisper, on-device AI, Hindi NLP	To approach

4.4 Success Metrics

100

BLIND USERS IN 6 MONTHS

1,000

PENDANTS SOLD IN YEAR 1

90%+

AI ACCURACY TARGET

3

NGO PARTNERSHIPS

Metric	3 Months	6 Months	12 Months
Blind assistive users	10 (beta)	100 (ADIP pilot)	3,000 (production)
BLE pendants sold	0	100 (early adopters)	1,000
Drone B2B licenses	0	5 (pilot)	50–200
Revenue	Rs 0 (build phase)	Rs 25L	Rs 1.5–4.2 Cr
AI accuracy (objects)	>85%	>90%	>95%
User satisfaction (NPS)	Baseline	>40	>60

The North Star

If a blind person in a tier-2 Indian city can walk to a grocery store, read product labels in Hindi, and pay — all guided by OmniPilot — we have won. Everything else is optimization.

OmniPilot Hardware

AI that sees, hears, and guides.

Soumya Swain

Co-founder | Sidewall

soumya@sidewall.in



Scan for LinkedIn — Kishore Rajendra

Sidewall Technologies Pvt. Ltd.

support@devpilot.co.in | devpilot.co.in

Confidential — April 2026